SUMMARY

In this chapter we have acquired the knowledge of

- Limit of a function $y = f(x)$ as x approaches x_0 from the lower values of x_0.
- Limit of $y = f(x)$ as x approaches x_0 from the higher values of x_0.
- Limit of a function as x approaches x_0, in the deleted neighbourhood of x_0 exists if and only if $\lim\limits_{x \to x_0^-} f(x) = L = \lim\limits_{x \to x_0^+} f(x) = \lim\limits_{x \to x_0} f(x) = L.$
- $\lim\limits_{x \to x_0} f(x) = L$ also means that $f(x)$ converges to L as x approaches x_0 from either side of x_0 except at $x = x_0$.
- If $\lim\limits_{x \to x_0} f(x)$ and $\lim\limits_{x \to x_0} g(x)$ exist then

  (i) $\lim\limits_{x \to x_0} [f(x) \pm g(x)] = \lim\limits_{x \to x_0} f(x) \pm \lim\limits_{x \to x_0} g(x)$

  (ii) $\lim\limits_{x \to x_0} [f(x).g(x)] = \lim\limits_{x \to x_0} f(x). \lim\limits_{x \to x_0} g(x)$

  (iii) $\lim\limits_{x \to x_0} \left[\dfrac{f(x)}{g(x)}\right] = \dfrac{\lim\limits_{x \to x_0} f(x)}{\lim\limits_{x \to x_0} g(x)}$ if $g(x) \neq 0$ and $\lim\limits_{x \to x_0} g(x) \neq 0$.
- Limit of $f(x)$ as x approaches x_0 does not exist if either $f(x) \to \pm\infty$ as $x \to x_0^-$ or $f(x) \to \pm\infty$ as $x \to x_0^+$ or $\lim\limits_{x \to x_0^-} f(x) = l_1 \neq l_2 = \lim\limits_{x \to x_0^+} f(x)$
- (M, ∞) is the neighborhood of $+\infty$, $M > 0$

  $(-\infty, K)$ is the neighborhood of $-\infty$, $K < 0$.
- If $f(x) \to \pm\infty$ as $x \to x_0$ then $x = x_0$ is a vertical asymptote.
- The line $y = l_1$ (or l_2) is a horizontal asymptote of the curve $y = f(x)$ if either $f(x) \to l_1$ as $x \to \infty$ or $f(x) \to l_2$ as $x \to -\infty$.
- $f(x)$ is continuous at x_0 if and only if

  (i) $\lim\limits_{x \to x_0} f(x) = f(x_0)$

  (ii) $\lim\limits_{\Delta x \to 0} [f(x_0 + \Delta x) - f(x_0)] = 0$

  (iii) $\lim\limits_{x \to x_0} f(x) = f\left(\lim\limits_{x \to x_0} x\right)$.
- Jump and removable discontinuities.

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